Fiscal Year:	FY 2020	**Task Last Updated:**	FY 05/05/2020
PI Name:	Seidler, Rachael D. Ph.D.		
Project Title:	Does Intermittent or Continuous Artificial Gravity Counteract Long Duration Bed Rest Induced Neurocognitive Declines?		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rachaelseidler@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-294-1722
Organization Name:	University of Florida		
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PI Web Page:			
City:	Gainesville	**State:**	FL
Zip Code:	32611-8205	**Congressional District:**	3
Comments:	NOTE: PI moved to University of Florida in July 2017; previous affiliation was University of Michigan.		
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-AGBR. Appendix G: Physiological & Behavioral Responses in Humans to Intermittent Artificial Gravity during Bed Rest
Start Date:	07/01/2018	**End Date:**	06/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Williams, Thomas	**Contact Phone:**	281-483-8773
Contact Email:	thomas.j.will1@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 8/28/20)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Mulavara, Ajitkumar Ph.D. (KBRWyle/NASA Johnson Space Center) Wood, Scott Ph.D. (NASA Johnson Space Center) Bloomberg, Jacob Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC18K0783		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA Research Announcement NNJ15ZSA001N-AGBR: Appendix G: Physiological and Behavioral Responses in Humans to Intermittent Artificial Gravity during Bed Rest invites proposals to investigate whether 30 minutes of artificial gravity (AG) per day, delivered either continuously or in six, five minute intermittent blocks, mitigates changes that occur in a spaceflight analog environment. We have recently completed a 70 day 6° head down tilt bed rest study in which we identified specific bed rest induced changes in brain structure, brain function, sensorimotor, and cognitive performance. Here, we select those that are sensitive to this spaceflight analog environment as outcome metrics for the following aims: Aim 1) Determine whether 30 minutes per day of supine centrifugation +1Gz at the heart is effective at mitigating neurocognitive declines induced by 60 days of 6° head down tilt bed rest in comparison to a control group that is not exposed to AG. Aim 2) Determine whether centrifugation is more effective and better tolerated as a countermeasure when it is delivered continuously for 30 minutes or in six, five minute sessions separated by breaks of five minutes each. Our approach utilizes cutting edge neuroimaging techniques and a broad ranging battery of sensory, motor, and cognitive assessments that we have found to be sensitive indices of bed rest—and in several cases—spaceflight induced performance decrements.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research on adaptive plasticity associated with behavioral training has greatly expanded over the past two decades, but research on maladaptive plasticity occurring with immobilization is scant. A greater understanding of brain structural and functional changes, and the concomitant behavioral effects resulting from limb disuse and unloading has implications for rehabilitation of those immobilized by injury, disease, or even simple inactivity.
Task Progress:	This project is currently being conducted in as a part of the “AGBRESA” (Artificial Gravity Bed Rest – European Space Agency) campaign at envihab in Cologne, Germany. It is a 60 day head down tilt bed rest study with daily centrifugation as a potential countermeasure. One group will not participate in centrifugation, one will receive 30 minutes per day of continuous centrifugation, and one will receive 30 minutes per day of intermittent centrifugation in 5 minute sessions. The original aims of the project are: Aim 1) Determine whether 30 minutes per day of supine centrifugation +1Gz at the heart is effective at mitigating declines in brain structure, brain function, sensorimotor and cognitive performance induced by 60 days of 6° head down tilt bed rest in comparison to a control group that is not exposed to AG. Aim 2) Determine whether centrifugation is more effective and better tolerated as a countermeasure when it is delivered continuously for 30 minutes or in six, five minute sessions separated by breaks of five minutes each. Aim 3) Identify predictors of individual differences in bed rest and AG responses. We will examine whether pre bed rest measures of brain structure and brain function are predictive of changes in sensory, motor, and cognitive function occurring in association with bed rest, and whether they also predict AG tolerance. We will combine data collected here with those we acquired for two previously National Space Biomedical Research Institute (NSBRI) funded “predictors” studies (Bloomberg, Mulavara, Principal Investigators PIs) to increase our predictive power. Since the initial funding of this award, we have achieved the following items: Recruited, interviewed and hired a postdoctoral fellow; Finalize pre, per, and post centrifugation assessments; Obtained NASA and University of Florida Institutional Review Board (IRB) approval; Trained German Aerospace Center (DLR) staff for data collection; Initiated an LSDA (Life sciences Data Archive) data submission agreement ; Reached a budget agreement; Met with Human Factors and Behavioral Performance (HFBP) contacts in which very minor project changes were approved; Traveled to envihab to set up the study, collect pilot data, and troubleshoot protocols; Data collection for all subjects was completed December, 2019; Data transfer has occurred and we are analyzing data.
Bibliography Type:	Description: (Last Updated: 03/12/2026)
Abstracts for Journals and Proceedings	McGregor HR, De Dios YE, Gadd NE, Koffman IS, Bloomberg JJ, Wood S, Mulvara AP, Seidler RD. "Effects of Continuous and Intermittent Artificial Gravity of Intracranial Free Water Distribution During Head-down Tilt Bed Rest." Podium presentation at the 2020 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 27-30, 2020. HRP IWS abstract booklet. 2020 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Abstracts for Journals and Proceedings	Tays GD, McGregor HR, Lee JK, Gadd N, Kofman IS, De Dios YE, Wood S, Mulder E, Bloomberg JJ, Mulavara AP, Seidler RD. "Does artificial gravity protect against the effects of a spaceflight analog environment on motor learning?" Society for the Neural Control of Movement Annual Conference, Dubrovnik, Croatia, April 27 – May 1, 2020 (was scheduled for these dates). This meeting was cancelled due to COVID-19. Society for the Neural Control of Movement 2020 conference abstract booklet. (in person meeting was cancelled due to COVID-19) , Mar-2020